

Quick § 8.1, 8.2, 8.3:

Let $F (= \mathbb{Q}, \mathbb{R}, \mathbb{C})$ be an ordered field and suppose we have $<, >, \leq, \geq$ behaving in ways we expect.

Theorem 6 (Trichotomy). If $x, y \in F$, then exactly one of the following occurs:

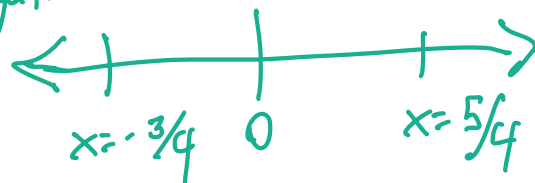
i) $x=y$, ii) $y < x$, or iii) $x < y$.

Jumping to § 8.2. The absolute value $|x|$ of $x \in F$ is defined to be $|x| \stackrel{\text{def}}{=} \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$.

"The distance from x to 0."

negative

Positive



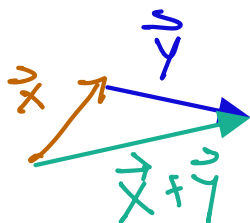
$$|\frac{5}{4}| = \frac{5}{4}$$

$$|x| = |-3/4| = -(-3/4) = 3/4$$

" $|x-y|$ the distance between x and y ."

Theorem 30. (The Triangle Inequality) If $x, y \in F$, then

$$|x+y| \leq |x| + |y|.$$



$$\|\vec{x}\| + \|\vec{y}\| \geq \|\vec{x} + \vec{y}\|$$

Proof. If $x \geq 0$, then $|x| = x$. If $x < 0$, then $|x| > x$.

Thus $|x| \geq x$. Hence $x+y \leq |x|+y \leq |x|+|y|$.

Likewise $-x-y \leq |x|-y \leq |x|+|y|$. If $x+y \geq 0$, then

$|x+y| = x+y \leq |x|+|y|$. If $x+y < 0$, then

$|x+y| = -x-y \leq |x|+|y|$. Therefore, $|x+y| \leq |x|+|y|$. $\checkmark$

Def. 7 (Intervals) For $x, y \in F$ with $x < y$ define $(x, y) \stackrel{\text{def}}{=} \{z \in F : x < z < y\}$.

For example, $(0, 1)$ in $\mathbb{R}$ is



Theorem 32 (Density) "Between any two rational numbers is another rational number" Let $x, y \in F$ with $x < y$. Then $\exists z \in F$ with $x < z < y$. In other words, $(x, y) \neq \emptyset$.

$$\frac{x+y}{2} = z.$$

$$\left(\frac{314159}{100000}, \frac{22}{7} \right)$$

$$\{z \in F : x < z < y\}$$

In $\mathbb{Z}$,

$$(1, 2) = \{z \in \mathbb{Z} : 1 < z < 2\} = \emptyset$$

Proof. Consider $z = \frac{x+y}{2} = 2^{-1}(x+y)$.

$$\frac{1+2}{2} = \frac{3}{2}$$

Since $x < y$, $x+x < x+y < y+y$. Thus $2x < x+y < 2y$.

As $2 \neq 0$, $2^{-1} = \frac{1}{2} \in F$. Since $2 > 0$, $\frac{1}{2} > 0$. Thus

$$\frac{1}{2} \cdot 2x < \frac{1}{2}(x+y) < \frac{1}{2} \cdot 2y. \text{ Hence } x < \frac{x+y}{2} < y$$

§ 8.3 The Archimedean Property.

Def. 10. F is Archimedean if $\mathbb{Q} \subset F$ as an ordered subfield and if $x, y \in F$ with $x > 0$, then $\exists n \in \mathbb{N} \subset F$ such that $nx \geq y$.

Choose $x=1$ to see that the natural numbers are unbounded in F .

Class. ^{really big #} 20

$\frac{1}{40}$

take any $n \in \mathbb{N}$
with $n \geq 20 \cdot 40$

$$\text{Then } n \cdot \frac{1}{40} \geq 20$$

Theorem 31. If F is an Archimedean ordered Field then $\mathbb{Q}$ is dense in F . In other words,

if $x < y$ in F , then there exists $r \in \mathbb{Q}$ such that
 $x < r < y$.

Ch. 8 §5: Infinite Sequences and limits

Def. An infinite sequence with values in F is a function whose domain is the natural numbers and whose codomain is F . Write a_i to denote the image of $i \in \mathbb{N}$ under this function.

we often augment/restrict this

Examples. $a_i = i$ is $\{0, 1, 2, 3, \dots\}$ in $\mathbb{Q}$

$a_i = \frac{1}{i}$ with $i > 0$ is $\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\}$

$$a_i = \frac{1}{i+1}$$

$b_i = \frac{-1}{i^2}$ with $i > 0$ is $\{-1, -\frac{1}{4}, -\frac{1}{9}, -\frac{1}{16}, \dots\}$

$c_i = (-1)^i \frac{1}{i^2}$ for $i > 1$ is $\{(-1)^2 \frac{1}{2^2} = \frac{1}{2}, -\frac{1}{6}, \frac{1}{12}, -\frac{1}{20}, \dots\}$

Problem 1: $\sqrt{2}$ and other friends are not in $\mathbb{Q}$

Solution: Take sequences of rationals

$$1, \frac{14}{10}, \frac{141}{100}, \frac{1414}{1000}, \frac{14142}{10000}, \dots \rightarrow \sqrt{2}$$

Problem 2: There are multiple sequences that approach the same thing

$$1, \frac{14}{10}, \frac{14}{10}, \frac{141}{100}, \frac{141}{100}, \frac{1414}{1000}, \frac{1414}{1000}, \frac{1414}{1000}, \frac{14142}{10000}, \dots \rightarrow \sqrt{2}$$

$$1, 1 + \frac{1}{3}, 1 + \frac{1}{3} + \frac{2}{27}, 1 + \frac{1}{3} + \frac{2}{27} + \frac{1}{243}, \dots \rightarrow \sqrt{2}$$

Solution: Equivalence classes of sequences

$$(a_i) \sim (b_i) \stackrel{(\text{def})}{\text{if}} \lim_{i \rightarrow \infty} a_i - b_i = 0$$

Problem 3: There are sequences that shouldn't converge

$$1, 2, 3, 4, 5, \dots$$

$$1, -1, 1, -1, 1, -1, \dots$$

Solution: Take only equivalence classes of sequences that "should" converge.

Define a sequence (a_i) to be Cauchy
if $\forall \varepsilon > 0 \exists N \in \mathbb{N}$ such that if $i, j \geq N$,
 $|a_i - a_j| < \varepsilon$.

Math Debt.

Prove sequences that converge are Cauchy.

Ch. 10, Defs 1 & 2 (The Real Numbers)

If (a_i) is a Cauchy sequence in $\mathbb{Q}$, then
define $[(a_i)]$ to be the equivalence class of
 (a_i) . I.e., $[(a_i)] = \{(b_i) : (b_i) \sim (a_i)\}$.

Define the real numbers, denoted by $\mathbb{R}$, to be
 $\mathbb{R} \stackrel{\text{def}}{=} \{[(a_i)] : (a_i) \text{ is a Cauchy sequence in } \mathbb{Q}\}.$

Def. 3 ($+$ & $\cdot$ on $\mathbb{R}$)

$$[(a_i)_{i \geq n_0}] + [(b_i)_{i \geq m_0}] = [(a_i + b_i)_{i \geq \max(n_0, m_0)}]$$

$$[(a_i)_{i \geq n_0}] \cdot [(b_i)_{i \geq m_0}] = [(a_i \cdot b_i)_{i \geq \max(n_0, m_0)}].$$

Do Operations with sequences work well?
 All from
 Chapter 8: Thm 44. Suppose $a_i = a$ for all i . Then $\lim_{i \rightarrow \infty} a_i = a$.
 $\frac{1}{2} = [(\frac{1}{2})]$

Thm 45. If $a_i \leq b_i$ for all i sufficiently large,
 then $\lim_{i \rightarrow \infty} a_i \leq \lim_{i \rightarrow \infty} b_i$.

Calc limit laws forever!

Define a sequence (a_i) to be Cauchy
 if $\forall \varepsilon > 0 \exists N \in \mathbb{N}$ such that if $i, j \geq N$,
 $|a_i - a_j| < \varepsilon$. sequences that should
 converge.

Ch. 9, Thm 15. Suppose $\lim_{i \rightarrow \infty} a_i = L$. Then (a_i) is
 Cauchy.

Proof. Let $\varepsilon > 0$. Since $\lim_{i \rightarrow \infty} a_i = L \exists N \in \mathbb{N}$

s.t. if $i \geq N$, then $|a_i - L| < \frac{\varepsilon}{2}$.

Now if $i, j \geq N$, then

$$|a_i - a_j| = |a_i - L + L - a_j| \leq |a_i - L| + |L - a_j| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

$\uparrow$ $+L-L$ trick $\uparrow$ Δ -inequality

Thus (a_i) is Cauchy. $\square$

Ch. 9
Thm. 16 If $(a_i) \cup (b_i)$ and (a_i) is Cauchy,
then (b_i) is Cauchy.

Scratch. Want $|b_i - b_j| < \varepsilon$

Have $|a_i - a_j|$ small and $|a_i - b_i|$ small

$$|b_i - b_j| = |b_i - a_i + a_i - a_j + a_j - b_j| \leq \underbrace{|b_i - a_i|}_{\frac{\varepsilon}{3}} + \underbrace{|a_i - a_j|}_{\frac{\varepsilon}{3}} + \underbrace{|a_j - b_j|}_{\frac{\varepsilon}{3}}$$

Proof. Let $\varepsilon > 0$. Since $(a_i) \cup (b_i)$ & (a_i) is Cauchy

$\exists N_1 \in \mathbb{N}$ s.t. if $i \geq N_1$, then $|a_i - b_i| < \frac{\varepsilon}{3}$ and

$\exists N_2 \in \mathbb{N}$ s.t. if $i, j \geq N_2$, then $|a_i - a_j| < \frac{\varepsilon}{3}$.

Let $N = \max(N_1, N_2)$. If $i, j \geq N$, then

$$|b_i - b_j| = |b_i - a_i + a_i - a_j + a_j - b_j| \leq |b_i - a_i| + |a_i - a_j| + |a_j - b_j|$$

$\uparrow$ Δ -inequality

$$< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

Thus (b_i) is Cauchy! $\square$

Final Big Idea: "The real numbers have enough stuff."
 $\mathbb{R}$ is Complete

Warm Up: Prove $\sqrt{2} \in \mathbb{R}$.

Step 1. Show the sequence a_i with $i \in \mathbb{N}$, defined by $a_i = \frac{d_0 \dots d_i}{10^i}$ with $\left(\frac{d_0 \dots d_i}{10^i}\right)^2 \leq 2$ but $\left(\frac{d_0 \dots d_i}{10^i} + \frac{1}{10^i}\right)^2 > 2$, $\checkmark$ is Cauchy.
here each d_j is in the range $0 \leq d_j \leq 9$

Step 2. Show $\lim_{i \rightarrow \infty} a_i^2 = 2$, so that $\sqrt{2} \in \mathbb{R}$ is $[(a_i)]$.

Want $|a_i - a_j| < \varepsilon$ for i, j sufficiently big. i, j zeros

$$\left| \frac{d_0 \dots d_i}{10^i} - \frac{d_0 \dots d_j}{10^j} \right| = \frac{d_0 \dots d_i}{10^i} - \frac{d_0 \dots d_j \underbrace{0 \dots 0}_{i-j \text{ zeros}}}{10^i} = \frac{d_{j+1} \dots d_i}{10^i}$$

$$< \frac{10^{i-j}}{10^i} = \frac{1}{10^j} \quad \leftarrow \text{choose } j \text{ big so } \frac{1}{10^j} < \varepsilon.$$

Proof of Step 1. Let $\varepsilon > 0$. Since $\mathbb{R}$ is Archimedean, $\exists n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$. Choose $N = n$ and suppose $i, j \geq N$. We have

assume $i > j$. If $i = j$, then $a_i - a_j = 0 < \varepsilon$.

If $j > i$, relabel

i, j zeros

$$\left| \frac{d_0 \dots d_i}{10^i} - \frac{d_0 \dots d_j}{10^j} \right| = \frac{d_0 \dots d_i}{10^i} - \frac{d_0 \dots d_j \underbrace{0 \dots 0}_{i-j \text{ zeros}}}{10^i}$$

$$= \frac{d_{j+1} \dots d_i}{10^i}$$

$$< \frac{10^{i-j}}{10^i} = \frac{1}{10^j} \leq \frac{1}{10^{n=N}} < \frac{1}{n} < \varepsilon.$$

Thus (a_i) is Cauchy, so

$$[(a_i)] \in \mathbb{R}.$$

Step 2 is for you!

Note $2 \in \mathbb{R}$ is the equivalence class of the constant sequence $(2)_{i \geq 0}$.

Ch. 8
Def. 14. (Supremum a.k.a. Least Upper Bound)

Let S be a non-empty subset of an ordered field F .

The Supremum M of S is the least upper bound:

$M \geq x \ \forall x \in S$ and if $B \in F$ is such that
 $B \geq x \ \forall x \in S$, then $M \leq B$.

The Infimum (greatest lower bound) has a similar definition.

Ch. 9, Def. 1. (Completeness) Let F be an ordered field.
We say F is complete if every nonempty subset S which is
banded from above has a supremum.

Ex. Consider $S \subset \mathbb{Q}$ defined by $S = \{r \in \mathbb{Q} : r^2 \leq 2\}$
It is banded above by 2. Does it have a supremum?
No, it having a Supremum is equivalent to $\sqrt{2} \in \mathbb{Q}$.

Ch. 8, Thm 6D a.k.a. Ch. 9, Thm 11. Let S
be a nonempty subset of an ordered Archimedean field F .
If M is the Supremum of S , then M is the limit of
a sequence (a_i) in S . Further, M is the limit of a

sequence (b_i) not in S . If $a < M$, we can assume each $a_i \in [a, M]$. If $b > M$, we can assume $b_i \in [M, b]$.

Punchline: Ch. 9, Thm. 18. (The Cauchy Criterion for Completeness)
Let F be an Archimedean ordered field.

If every Cauchy sequence in F converges, then F is complete.

This will end up showing that $\mathbb{R}$ is complete.

Ch. 9, Def. 2 (Continuity). " f is a continuous function if you can graph it without picking up your pencil."

Let S be a subset of an ordered field F . Then a function

$f: S \rightarrow F$ is said to be continuous at S if for all convergent sequences (a_i) with terms and limit in S , the sequence $(f(a_i))$ also converges and $\lim_{i \rightarrow \infty} f(a_i) = f(\lim_{i \rightarrow \infty} a_i)$. " f commutes with limits."

" f respects convergent sequences"

"A convergent sequence in the domain becomes a convergent sequence in the codomain."

Ex.

$f: \mathbb{Q} \rightarrow \mathbb{Q}$ given by $f(r) = r$.

The identity function is continuous.

$g(x) = x^2$ as a function from $\mathbb{Q}$ to $\mathbb{Q}$ is continuous.

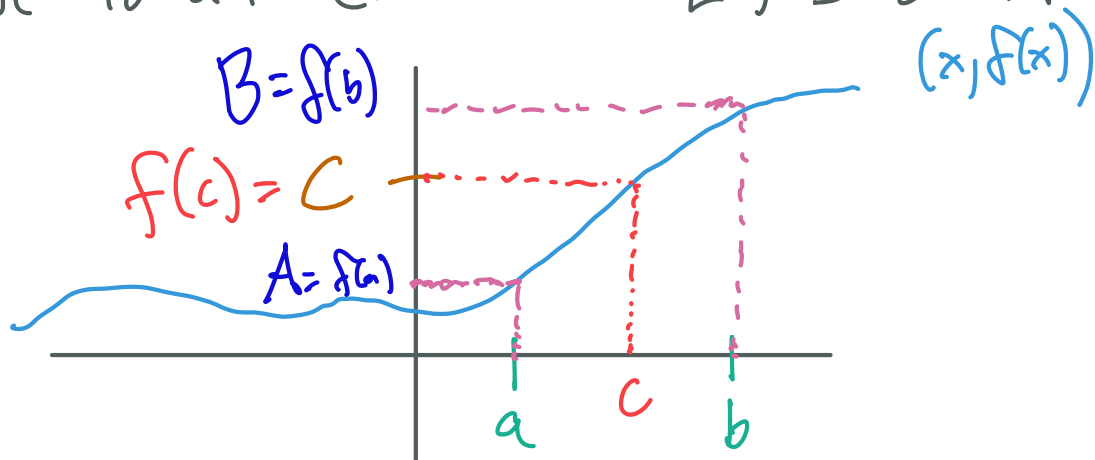
$h(x) = \sqrt{x}$ as a function $\mathbb{R}^{\geq 0} \rightarrow \mathbb{R}^{\geq 0}$ is continuous

However this function is not defined on $\mathbb{Q}$.

Ch. 9, Thm. 12 (Intermediate Value Theorem)

Let F be a complete ordered field that contains $\mathbb{Q}$ as an ordered subfield. Let $[a, b] \subset F$ where $a < b$.

Suppose $f: [a, b] \rightarrow F$ is continuous. If $C \in F$ is any value between $A = f(a)$ and $B = f(b)$, then there is an element $c \in [a, b]$ such that $f(c) = C$.

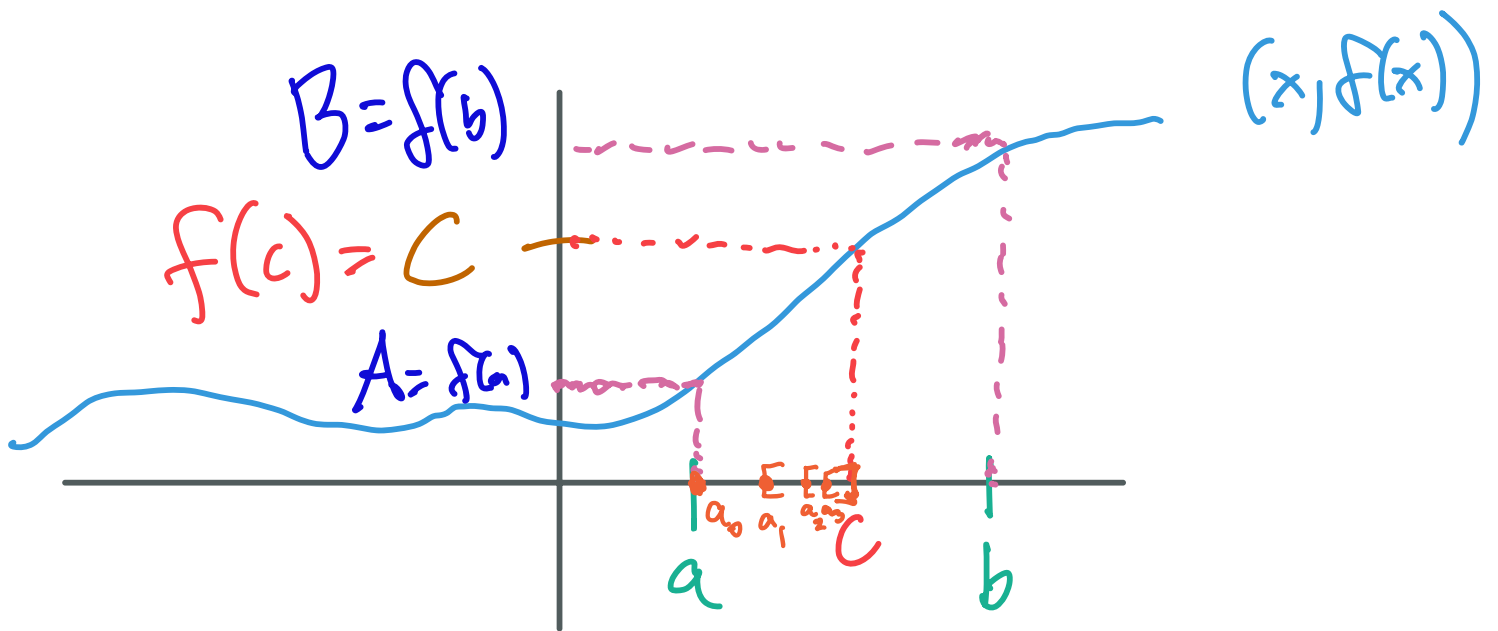


Proof. WLOG assume $A = f(a) < C < B = f(b)$.
 (note if $C = A$, then $f(a) = C$ and if $C = B$, then $f(b) = C$)

Define $S = \{u \in [a, b] : f(u) \leq C\}$. Note S is nonempty

Since $a \in S$ and b is an upper bound for S . Since F is complete, S has a supremum (least upper bound) c .

We begin by showing $f(c) \leq C$. By Thm. 11 $\exists$ a sequence (a_i) in S converging to c . We construct (a_i) in the following manner, $a_0 = a$. For each $i \in \mathbb{N}$, choose $a_{i+1} \in [\frac{a_i + c}{2}, c] \cap S$. Such a choice is possible since c is the least upper bound.



We see $|c - a_i| = c - a_i \leq \frac{c - a}{2^i}$. Fix $\varepsilon > 0$.

$$c - a_2 \leq c - \frac{c + a_1}{2} = \frac{2c}{2} - \frac{c + a_1}{2} = \frac{c - a_1}{2} \leq \frac{c - \frac{a + c}{2}}{2} = \frac{\frac{c - a}{2}}{2} = \frac{c - a}{2^2}$$

Since we're Archimedean, $\exists n \in \mathbb{N}$ s.t. $\frac{1}{n} < \varepsilon$ $\exists m \in \mathbb{N}$ s.t. $m > c - a$.

Let $N = 2^{\max(m, n)}$. If $i \geq N$, then

$$|C - a_i| \leq \frac{C - a}{2^i} \leq \frac{C - a}{2^N} \leq \frac{C - a}{2^m} \cdot \frac{1}{2^n} \leq \frac{C - a}{m} \cdot \frac{1}{n} < 1 \cdot \frac{1}{n} < \varepsilon.$$

Thus $\lim_{i \rightarrow \infty} a_i = C$. Since each $a_i \in S$, we have $f(a_i) \leq C$.

Since f is continuous $f(C) = \lim_{i \rightarrow \infty} f(a_i) \leq C$
halfway to showing $f(C) = C$.

Next note $C \in [a, b)$, b is an upper bound for S and C is the least upper bound, so $C \leq b$ but

$$f(b) = B > C \geq f(C), \text{ so } C \neq b.$$

Lastly we'll show that $f(C) \geq C$. By Thm. 11 there is a sequence (b_i) of elements of $[C, b]$ not in S converging to C .

$$\text{That is, } \lim_{i \rightarrow \infty} b_i = C$$

Now since $b_i \notin S$, we see $f(b_i) \geq C$. Further, f is continuous so $f(C) = f(\lim_{i \rightarrow \infty} b_i) = \lim_{i \rightarrow \infty} f(b_i) \geq C$.

Thus $f(C) \geq C$. Hence $f(C) = C$.

Cor. one can show $f(x) = x^2$ is continuous.

We see $f(1) = 1^2 = 1$ & $f(2) = 2^2 = 4$. Thus

if $\mathbb{Q}$ was complete there would be a $c \in [1, 2]$

such that $f(c) = 2 \in [f(1), f(2)] = [1, 4]$.

Hence $\mathbb{Q}$ is not complete.

Decimal Expansions

Ex. $\frac{2}{3}$ as a decimal

$0.666\dots$

$= 0.\overline{6}$

indicates the repeating 6.

$3 \overline{) 2.00}$

$\underline{1.8}$

0.20

$\underline{0.18}$

0.02

More theoretically, we are writing

$$\frac{2}{3} \text{ as } n + \sum_{j=1}^{\infty} \frac{d_j}{10^j}$$

$$\frac{2}{3} = [(a_i)]$$

where a_i is the i th partial sum

$$a_i = n + \sum_{j=1}^i \frac{d_j}{10^j} = n + \frac{d_1 \dots d_i}{10^i}$$